Cross-Lingual Legal Summarization

1. The Problem

Can prompting strategies improve Hindi legal summarization by small local language models?

- Legal judgments are long, complex, and difficult to summarize accurately.
- The task is to generate concise Hindi summaries that preserve the important legal information and final ruling.
- This experiment evaluates three prompting strategies:
  - Zero-shot
  - Chain-of-Thought (CoT)
  - Few-shot

Difficulties:

English-input → Hindi-output legal summarization is difficult because it combines four challenges: cross-lingual linguistic divergence, specialized legal terminology, long-document compression, and faithfulness. English–Hindi structural differences require substantial reordering, morphology, and gender/register decisions. Legal judgments contain archaic constructions, Latin maxims, and terms without precise Hindi equivalents. Long judgments also suffer from positional bias, causing important reasoning to be missed. Most critically, hallucination and mistranslation can silently alter legal meaning.

2. Dataset

SOURCE	*https://github.com/Law-AI/MILDSum/tree/main/Data/MILDSum_Samples*
NUMBER OF ENTRIES	*10*
TASK	*Legal judgment → concise Hindi summary*
REFERENCE	*HINDI SUMMARY PROVIDED IN DATASET*
INPUT	*ENGLISH JUDGEMENT TEXT*
OUTPUT	*HINDI SUMMARY GENERATED BY MODEL*

EN_Judgment.txt

1. The petitioner, who is a serving officer of the Delhi Higher Judicial Services, and is currently posted as the Additional District Judge- 02 South District Saket Courts New Delhi, has approached this Court being aggrieved
by the refusal of the respondent nos. 1 to 3 in reimbursing in full the expenses incurred by him for his medical treatment, while he was admitted at the respondent no.5/hospital, between 22.04.2021 to 07.06.2021, on account of Covid-19.
2. Learned senior counsel for the petitioner submits that the respondent nos.1 to 3 do not dispute the fact that the petitioner was undergoing....

EN_Summary.txt

The Delhi High Court on Tuesday directed the Delhi Government to pay over Rs. 1>
ADJ Dinesh Kumar was admitted in the city's PSRI Hospital between April 22 to J>
Justice Rekha Palli said undoubtedly the authorities are justified in urging th>
"However, the fact remains that during April and May, 2021, when the residents >
Allowing the plea, the court observed that the judge, who had to spend his hard>
"This Court does not deem it appropriate or necessary to delve into the validit>
Justice Palli however rejected Delhi Government's....

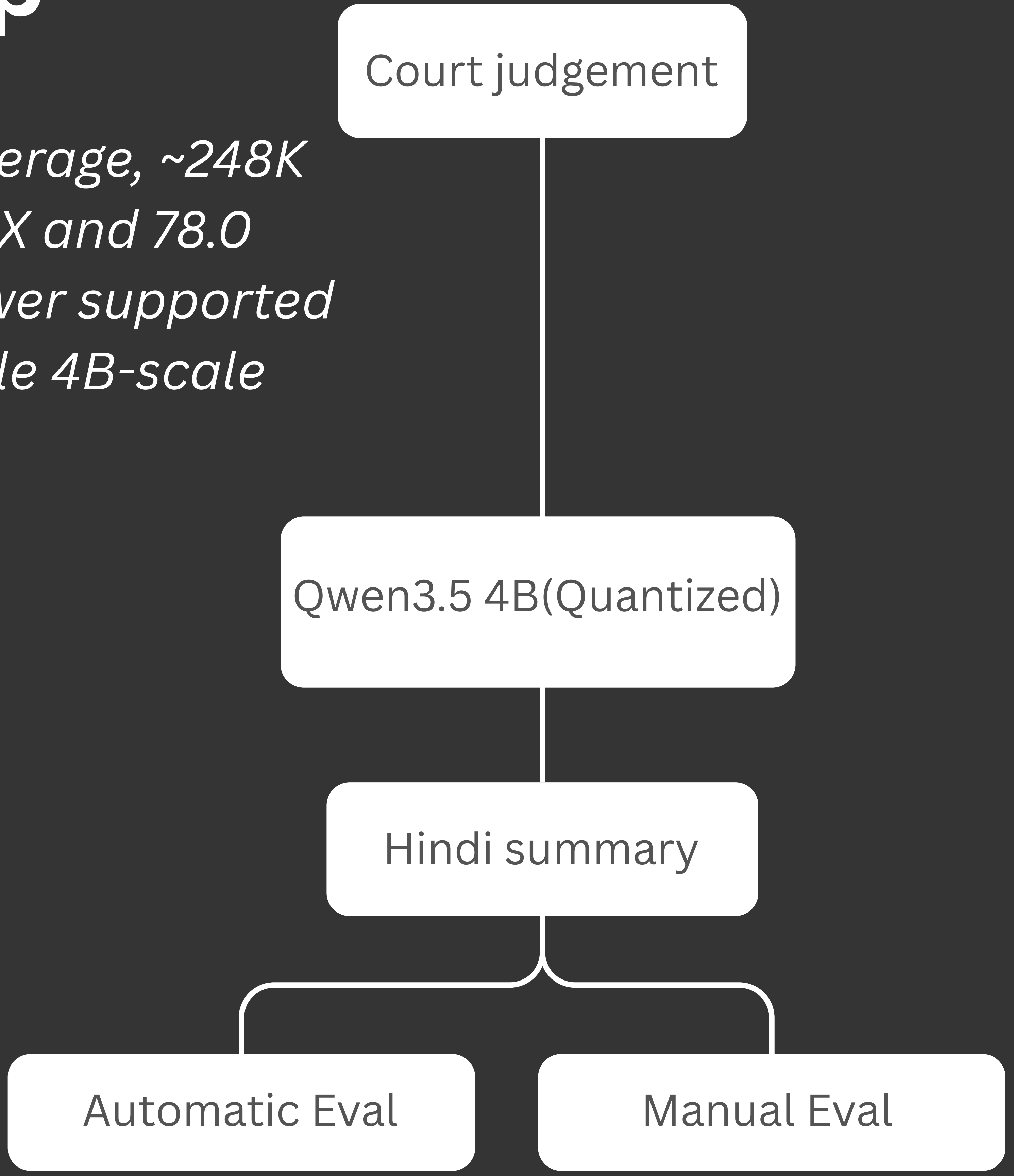
HI_Summary.txt

दिल्ली हाईकोर्ट (Delhi High Court) ने मंगलवार को दिल्ली सरकार को एक वरिष्ठ न्यायिक अधि>
एडीजे दिनेश कुमार को COVID-19 की दूसरी लहर के दौरान 22 अप्रैल से 7 जून, 2021 के बीच शहर के>
जस्टिस रेखा पल्ली ने कहा कि निस्संदेह अधिकारियों का यह आग्रह करना उचित है कि अस्पताल ने जीए>
अदालत ने कहा,
"तथ्य यह है कि अप्रैल और मई, 2021 के दौरान, जब दिल्ली के निवासी न केवल अस्पताल के बिस्तर प>
याचिका को स्वीकार करते हुए, अदालत ने कहा कि न्यायाधीश, जिन्हें अपनी जान बचाने के लिए COVI>
कोर्ट ने कहा...

3. Model and Experimental Setup

Model: Qwen3.5 4B Q4_k_m

The reason why I chose Qwen3.5-4B is its 201-language coverage, ~248K vocabulary, and strong multilingual results: 71.5 MMLU-ProX and 78.0 MAXIFE. While Llama has weaker Indic tokenization and fewer supported languages; Gemma 4 is competitive but has less comparable 4B-scale Hindi evidence.



EXPERIMENTAL SETUP:

Dataset: 10 Indian court judgements

Task: Generate concise Hindi legal summaries

Prompting strategies:

Zero-Shot	**Few-Shot**	**COT(Chain of thought)**

The same 10 judgments were used across all experiments
Inference and evaluation conditions were kept consistent
Outputs compared against reference summaries.

Objective:

Evaluate whether prompting strategy changes summary quality when using a small LLM on real Indian legal judgments and understand the limitations of automatic evaluation methods.

4. Three Prompting Strategies

English Input

Prompt(Zero-Shot):

You are an expert legal journalist. Convert the following court judgment into a concise, readable news summary. The summary must capture the essence of the judgment who, what, when, where, why, and the final ruling in a continuous flowing narrative

Follow these rules strictly:

1. Lead with the key ruling: Start with the court name, date, and the core direction or decision in one powerful opening sentence.
2. Maintain chronological flow: Present facts, arguments, and the final order in the sequence they appear in the judgment.
3. Include critical quotes: Incorporate 2-3 essential quotes from the judgment verbatim, enclosed in double quotation marks.
4. Preserve all parties and amounts: Mention the petitioner, respondent(s), and any specific monetary amounts or orders exactly as stated.
5. End with case metadata: Conclude with the case title, bench name, and counsel names exactly as they appear.
6. No editorializing: Do not add opinions, interpretations, or commentary. Stick strictly to what the judgment states.
7. Continuous text: Write as a cohesive news article — no bullet points, no markdown formatting, no section headers. Just flowing paragraphs.

Judgment Text: Context**

Summary:

Hindi Output

English Input

Prompt:(Few-Shot)

You are an expert legal journalist. Convert the following court judgments into concise, readable news summaries in HINDI. The summary must capture the essence of the judgment — who, what, when, where, why, and the final ruling — in a continuous flowing narrative.

Follow these rules strictly:
1. Lead with the key ruling: Start with the court name, date, and the core direction or decision in one powerful opening sentence (in Hindi).
2. Maintain chronological flow: Present facts, arguments, and the final order in the sequence they appear in the judgment.
3. Include critical quotes: Incorporate 2-3 essential quotes from the judgment verbatim, enclosed in double quotation marks.
4. Preserve all parties and amounts: Mention the petitioner, respondent(s), and any specific monetary amounts or orders exactly as stated.
5. End with case metadata: Conclude with the case title, bench name, and counsel names exactly as they appear.
6. No editorializing: Do not add opinions, interpretations, or commentary. Stick strictly to what the judgment states.
7. Continuous text: Write as a cohesive news article — no bullet points, no markdown formatting, no section headers. Just flowing paragraphs.

Example 1

Judgment Text: 1. The petitioner, Ramesh Singh, has filed the present writ petition challenging the termination order dated 12.05.2021 passed by the Municipal Corporation of Delhi (Respondent No. 2). 2. The petitioner was appointed as a Junior Clerk in 2010. He was terminated on grounds of unauthorized absence from duty. 3. Learned counsel for the petitioner argued that the petitioner......

Hindi Output

English Input

Prompt:(CoT)

You are an expert legal journalist. Convert the following court judgment into a concise, readable news summary in HINDI. The summary must capture the essence of the judgment — who, what, when, where, why, and the final ruling — in a continuous flowing narrative.

Follow these rules strictly for the final summary:
1. Lead with the key ruling: Start with the court name, date, and the core direction or decision in one powerful opening sentence (in Hindi).
2. Maintain chronological flow: Present facts, arguments, and the final order in the sequence they appear in the judgment.
3. Include critical quotes: Incorporate 2-3 essential quotes from the judgment verbatim, enclosed in double quotation marks.
4. Preserve all parties and amounts: Mention the petitioner, respondent(s), and any specific monetary amounts or orders exactly as stated.
5. End with case metadata: Conclude with the case title, bench name, and counsel names exactly as they appear.
6. No editorializing: Do not add opinions, interpretations, or commentary. Stick strictly to what the judgment states.
7. Continuous text: Write as a cohesive news article — no bullet points, no markdown formatting, no section headers. Just flowing paragraphs.

Before writing the final summary, you must think step-by-step. In your thinking process:
1. Extract the Court name, Date, Case Title, Bench, and Counsel names.
2. Identify the Petitioner, Respondent, and any specific monetary amounts or dates.....

Hindi Output

5. **Automatic Evaluation**

The models were evaluated on 10 MILDSum samples using ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore (MuRIL). ROUGE was calculated using a Hindi-compatible tokenizer without stemming.
MethodROUGE-1ROUGE-2ROUGE-LBERTScore F1

Methods	ROUGE-1	ROUGE-2	RUUGE-L	BERT
Zero-Shot	0.3758	0.1157	0.1591	0.9737
Few-Shot	0.3923	0.1197	0.1615	0.9740
CoT	0.3839	0.1114	0.1568	0.9732

The results show that few-shot prompting performed best overall. It achieved the highest score on all three ROUGE metrics and the highest BERTScore F1, although the differences in BERTScore were very small. A notable finding is that CoT did not improve performance. Despite having the highest variability across samples, its average ROUGE-2, ROUGE-L, and BERTScore F1 were lower than both zero-shot and few-shot prompting.

KEY TAKEWAY: *Few-shot prompting is the strongest configuration according to automatic metrics, while CoT provides no measurable advantage and is less consistent.*

6. Manual Evaluation

Criteria	Zero-Shot	Few-Shot	CoT
Factuality	1.7	1.5	1.5
Coverage	2.6	2.8	3.1
Legal Correctness	2.1	1.7	1.9
Faithfulness	1.5	1.4	1.3
Hindi Quality	1.9	1.8	1.7
Overall Average	1.92	1.84	1.92

KEY TAKEWAY: All three methods perform poorly (≈1.8–1.9 out of 5). CoT has the best coverage but the worst faithfulness, while Zero-shot and CoT tie on overall average. Few-shot underperforms slightly on factuality and legal correctness.

7. Error Analysis

Error Type	Zero-Shot	Few-Shot	CoT
Hallucination	3.9	4.1	4.7
Important Omissions	5.0	4.5	4.4
Wrong Facts	5.3	5.0	5.9
Wrong Legal Conslusions	1.8	1.9	2.1
Translation Errors	4.4	4.3	4.8
Incorrect Legal Terminology	3.9	3.6	4.7

Common Errors in Generated Summaries

*All three methods produce fundamentally unreliable legal summaries, averaging 4-5 serious errors per case. The most dangerous error pattern is **inverted legal reasoning**, where summaries state the opposite of the judgment's actual holding such as concluding an officer "should bear excess charges" when the court held the employer must pay, or presenting a bail rejection as an "arrest" or "conviction." This mischaracterization of proceedings appears consistently across all methods and could seriously mislead legal practitioners. **Court and judge identification** is another systematic failure: Bombay High Court becomes "Maharashtra High Court," J&K High Court becomes "Uttar Pradesh High Court," single judges become division benches, and judges' names are omitted in 9 out of 10 rounds despite being clearly stated in the source. **Party confusion** is rampant counsel names are routinely misidentified as petitioners, and companies are transformed into individuals, such as "M/s Shree Guru Kripa Alloys" becoming the fabricated "Mashrook Ali Khan." **Translation errors** severely degrade comprehension: "bail" becomes "बेल" or "बॉल" instead of "जमानत," "reimbursement" becomes "रिसाव" (leakage), "circular" becomes "सैक्युलर" (secular), and "Standing Order" becomes "तालाब" (pond). CoT produces the most content but also the most hallucinations and wrong facts, while Zero-shot omits the most information and Few-shot offers the best balance yet still remains deeply flawed. Critical legal doctrines are consistently omitted Article 21 reproductive rights, retrospective operation of executive orders, Section 37 twin conditions, and the triple identity test in trademark cases all appear in fewer than half the summaries. **Mixed scripts** (Cyrillic, Chinese, and Latin characters appearing in Devanagari text) and untranslated English passages further render many summaries incomprehensible to Hindi readers.*

8. Metrics VS Manual Evaluation

1. *Does the method with the highest ROUGE/BERTScore also look best manually?*

→ **No**. Few-shot achieves the highest score on all four automatic metrics, but it does not perform best under manual evaluation. Zero-shot and CoT both receive the highest manual average (1.92), compared with 1.84 for Few-shot. This suggests that similarity to the reference is not the same as summary quality. Automatic metrics reward lexical/semantic overlap, while manual evaluation also considers whether the summary is factually and legally correct.

2. *Are there cases where the metric gives a good score but the summary is actually wrong?*

→ **Yes**, several,

- Sample 2: **CoT** gets ROUGE-1 0.453 and BERTScore 0.975, yet it fabricates the petitioner's name and gives the wrong court.
- Sample 4: **Few-shot** gets ROUGE-1 0.401 and BERTScore 0.973, but misrepresents a bail rejection as arrest/conviction.
- Sample 8: **CoT** gets ROUGE-1 0.428 and BERTScore 0.974, despite translating "Standing Order" as "तालाब" (pond) and giving the wrong bail amount.

So, a high metric score can coexist with a serious legal or factual error.

3. *Are there cases where the metric gives a low score but the summary is actually good?*
→ **Yes,** the clearest example is Sample 3 with **CoT**. It has the lowest ROUGE-1 score (0.274) among the three methods, yet has the highest manual score (1.8) for that case because it provides better factual coverage and captures important legal details. Similarly, in Sample 9, Few-shot has a relatively lower ROUGE-1 (0.377) but has the highest manual score of 3.0, with evaluators noting its strong coverage and detail.

Overall conclusion:
The comparison shows that ROUGE and BERTScore are poor proxies for legal-summary quality in this task. They can overestimate incorrect summaries and underestimate good, differently worded summaries. **Therefore, automatic metrics should be used as supporting measures, not as a replacement for manual evaluation, especially when factual and legal correctness are critical.**

9. Interesting Cases

Case 1: Few-shot Clearly Outperforms

Sample 9 shows a clear advantage for Few-shot prompting. It correctly translates the legal maxim that “justice must not only be done but must also appear to have been done,” identifies Aditi Bakht as the mother and petitioner, and captures important case details such as the guardianship petition and the child’s identity. Its manual score of 3.0 was the highest, despite having a lower ROUGE-1 score than Zero-shot and CoT. Zero-shot introduced errors such as “संयुक्त न्याय” and “पक्षी,” while CoT incorrectly referred to the child as a son and mistranslated “Child Counselor.” This case demonstrates that legal accuracy and appropriate terminology can matter more than lexical overlap with a reference summary.

Case 2: Catastrophic CoT Translation Error

Sample 8 demonstrates how a seemingly reasonable summary can become legally unusable because of a few critical translation errors. Zero-shot correctly rendered “Standing Order 1/88” as “स्थायी आदेश 1/88”, while Few-shot used the less standard “स्थापना नियम.” CoT produced catastrophic mistranslations: “Standing Order” became “तालाब” (pond), “Union of India” became “संयुक्त राज्य अमेरिका” (USA), and “Section 37” became “Article 37.” It also reported an incorrect bail amount. Consequently, CoT received the lowest manual score (1.6), even though BERTScore remained almost identical across all three methods (0.974–0.976). The case shows that automatic semantic similarity can fail to detect legally destructive terminology errors.

Case 3: Automatic Metrics vs. Manual Evaluation

Sample 2 highlights the strongest mismatch between automatic metrics and human judgment. All three systems incorrectly identified the court, but Few-shot introduced an especially serious error by reversing the direction of the furnace conversion and inventing a date. It achieved the highest BERT-F1 (0.978) but received the lowest manual score (1.4). CoT, despite fabricating the petitioner’s name as “Mashrook Ali Khan,” correctly captured more of the legal reasoning and achieved the highest manual score (1.8), while receiving a slightly lower BERT-F1 (0.975). This suggests that named-entity hallucinations, factual inversions, and legal reasoning errors are not adequately reflected in similaritybased metrics.

Conclusion

These cases show that automatic metrics alone are unreliable for evaluating Hindi legal summarization. ROUGE and BERTScore can reward lexical or semantic similarity while overlooking hallucinated entities, incorrect legal terminology, factual inversions, and even catastrophic mistranslations. Conversely, summaries with better legal terminology and reasoning may receive lower automatic scores. Manual evaluation is therefore essential for identifying errors that matter most in legal settings.

10. Key Findings

Key Findings: Hindi Legal Summarization Evaluation

This evaluation of Hindi legal summarization reveals a troubling disconnect: the methods that score highest on automatic metrics consistently produce the most unreliable summaries, while the legally best summaries are penalized for using proper Hindi terminology.

Across 10 legal judgments from the MILDSum dataset, three prompting strategies i.e. zero-shot, few-shot, and chain-of-thought were evaluated using ROUGE, BERTScore, and manual assessment by legal experts. The findings are stark.

Few-shot achieved the highest ROUGE-1 (0.392) and BERTScore F1 (0.974), yet scored lowest on manual evaluation (1.84 out of 5). Conversely, CoT and zero-shot tied for the best manual scores (1.92) despite lower automatic scores.

This inversion is not incidental. The reason is clear: metrics measure surface-level similarity, not legal accuracy. They cannot detect fabricated party names, wrong court identifications, or legal inversions. In one case, CoT translated "Standing Order" as "तालाब" (pond) and "Union of India" as "United States of America"—catastrophic errors that BERTScore treated as equivalent to correct translations.

The manual analysis identified systematic failures: judge names omitted in 9/10 rounds, counsel names in 7/10, and key legal doctrines like Article 21 reproductive rights and Section 37 twin conditions consistently missing.

Hallucinations were rampant CoT produced the most (46), followed by zero-shot and few-shot (40 each). Translation errors plagued all methods, with "bail" rendered as "बेल" instead of "जमानत" and "reimbursement" as "रिसाव" (leakage).

The implications are serious. Legal practitioners relying on these summaries would receive misleading information bail rejections characterized as arrests, employer obligations inverted to officer liability, and critical legal tests rendered incomprehensible. ROUGE and BERTScore cannot be trusted as primary evaluation tools for legal summarization in Hindi. Expert manual evaluation, task-specific error detection, and source verification are essential for any reliable assessment framework.

11. What to do Next?

Limitations:

➜ Sample size is extremely small so findings might not generalize well.
➜ Manual evaluation is subjective and may vary across samples.
➜ Human evaluation can be prone to errors on larger examples.
➜ Reference Based automatic evaluation penalizes. Human evaluators can have motivation problem leading to errors or “vibe” based numbering.
➜ Reference Based automatic evaluation penalizes valid output phrases.

Future Direction:

➜ Develop domain specific metrics.
➜ Build larger database with more varies examples.
➜ Using a specialized/fine-tuned model might yield better results.
➜ Develop domain specific metrics. Developing a Human-in-the-loop like hybrid evalutaion system